



Digital IC Design & Verification
Physical Design
Lab Manual # 02 | Synthesis of RISC-V Single Cycle Processor

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NUST Chip Design Centre (NCDC), Islamabad, Pakistan

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Revision History

Revision Number	Revision Date	Revision By	Nature of Revision	Approved By
1.0	02/05/2025	Hira Sohail	Lab Manual	Sharjeel Khilji
2.0	30/09/2025	Fawad Ahmed	Additional Tasks	Sharjeel Khilji



Lab Task

Perform complete RTL-to-gate-level Physical-Aware synthesis of your RISC-V Single Cycle Core (developed in the CA module). Make sure that in RISC-V, don't include IMEM & DMEM Modules in the Core. The task includes the following steps:

1. Ensure RTL is Synthesizable

- Modify your RTL code if necessary to make it fully synthesizable.
- Avoid synthesizable constructs and verify.

2. .sdc and script file

- Write appropriate constraint file and script to run synthesis.

3. Perform RTL Synthesis

- Run synthesis with all advance optimizations.
- Use appropriate constraints.

4. Timing Optimization

- Optimize your design to run at the highest possible frequency without timing violations.
- Report maximum achievable frequency and slack of critical path.

5. Post-Synthesis Simulation

- Perform gate-level simulation using the synthesized netlist.
- Compare functional behavior and timing of post-synthesis simulation with your original RTL simulation.

6. Detailed Report

- All synthesis steps and tool commands
- Design constraints and setup
- Area, power, and timing results
- Issues encountered and how they were resolved
- Simulation comparison with waveform snapshots (RTL vs post-synthesis).